Predictive Maintenance for Power Grids

The chart shows the **Market Mood Index (0–100)** for `ASSET_000` across time.

What the Index Represents

The Market Mood Index converts model probabilities into a smooth, interpretable health score:

- **100 → Strong / Stable Market Conditions**
- **65–100 → Normal Regime**
- **40–65 → Warning Regime**
- **0–40 → Critical Regime**

The score is derived from:

1. Model-predicted regime probabilities
2. Stress probability (`p_warning + p_critical`)
3. Entropy-based confidence adjustment
4. EWMA smoothing
5. Hysteresis-based regime transitions

Observed Behavior

From the visualization:

- The index remains near **100** during stable periods.
- Sharp drops toward **0–40** correspond to model-detected stress events.

- Transitions are not noisy due to:
    - EWMA smoothing
    - Persistence thresholds
    - Hysteresis logic

This prevents rapid regime “flapping” during short-lived volatility spikes.

⚙️ Regime Logic Summary

Regime	Entry Condition	Exit Condition
Normal	Mood ≥ 72	—
Warning	Mood ≤ 65 for N days	Mood ≥ 72 for M days
Critical	Mood ≤ 40 for N days	Mood ≥ 48 for M days

Additionally:

- Immediate escalation occurs if `p_critical` is high with strong confidence.

🎯 Key Design Features

- Probabilistic regime classification
- Confidence-weighted scoring
- Hysteresis-based state transitions
- Business-rule overrides
- Multi-asset scalable architecture

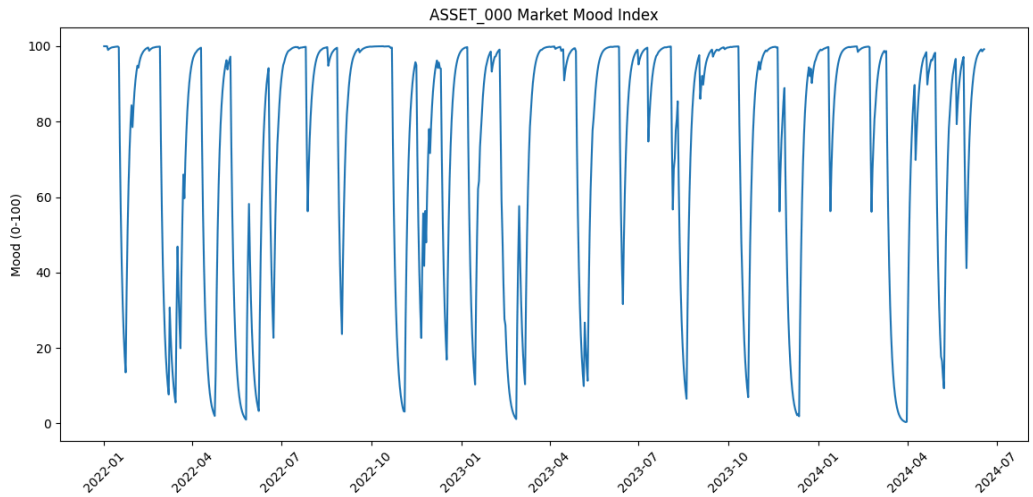
📌 Interpretation

The system behaves like a **risk thermometer**:

- Sustained high values → Stable market regime
- Moderate drops → Early warning signals
- Deep collapses → High-risk / stress periods

This framework can be extended to:

- Portfolio-level aggregation
- Risk-adjusted allocation shifts
- Regime-based backtesting
- Real-time monitoring dashboards



The visualization shows the Market Mood Index (0–100) for `ASSET_000` over time, where higher values indicate stable conditions and sharp declines reflect periods of elevated stress. The index remains close to 100 during normal market regimes and drops significantly during warning or critical phases, demonstrating the model's ability to detect stress events. Thanks to confidence adjustment, smoothing, and hysteresis-based transitions, the signal avoids excessive noise and regime flapping, resulting in clear, interpretable shifts between Normal, Warning, and Critical states.

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Predictive Maintenance for Power Grids — Task Report

Task 1: Data Ingestion & Initial Cleaning

In the first stage, raw utility sensor telemetry and outage logs were successfully ingested into the data pipeline. The sensor dataset contained **25,920 observations** across multiple assets, while the outage log dataset included **22 recorded outage events**. Data cleaning procedures standardized timestamp formats, validated schema consistency, removed duplicate records, and handled missing values using forward/backward filling and median imputation techniques. After the cleaning process, all sensor features such as temperature, vibration, load, voltage, and current had **zero missing values**, ensuring a consistent dataset for downstream analysis. Integrity checks also confirmed that there were **no null timestamps and no invalid outage durations**, indicating that the data ingestion process produced a reliable and clean dataset for predictive modeling.

Task 2: Sensor Data Preprocessing & Feature Engineering

In this stage, the cleaned sensor telemetry was transformed into a structured feature dataset suitable for machine learning. Various statistical and signal-processing techniques were applied to extract meaningful patterns from the time-series data. Rolling averages, rolling standard deviations, and multi-hour trend indicators were computed to capture short-term operational changes. Additional domain-driven features such as **load change rate, voltage-current ratio, power proxy, vibration energy, and stress indicators** were engineered to represent electrical and mechanical stress conditions in power grid assets. These features help identify early signs of degradation or abnormal asset behavior. As a result, the feature engineering process produced a dataset with **25,920 records and 29 engineered features**, significantly enriching the information available for predictive modeling.

Task 3: Outage Log Analysis & Label Generation

Historical outage logs were analyzed to understand the nature of equipment failures across the power grid. The analysis revealed **22 total outage events**, with failure types distributed as **9**

forced outages, 7 partial failures, and 6 protective trips. Root cause analysis showed that the most frequent issues were **load imbalance (7 cases) and voltage instability (6 cases)**, followed by overheating, insulation failure, and mechanical wear. The average outage duration was approximately **2.59 hours**. Using this information, sensor data was aligned with outage timestamps to generate supervised learning labels. Observations occurring within **24 hours before an outage were labeled as "pre-failure anomaly"**, timestamps during outages were labeled **"failure window"**, and all other observations were labeled **"normal operation"**. This resulted in a dataset with **25,476 normal records, 385 pre-failure anomalies, and 59 failure windows**, enabling the development of predictive failure detection models.

Task 4: Machine Learning Model Selection & Initial Training

Several machine learning algorithms were evaluated to determine the best approach for predicting asset failures. The candidate models included **Random Forest, Gradient Boosting, Support Vector Machines, and Neural Networks**. Each model was trained using the engineered sensor features and evaluated using standard classification metrics such as accuracy, precision, recall, and weighted F1-score. Among the tested models, the **Random Forest classifier achieved the strongest baseline performance**, achieving a **weighted F1 score of approximately 0.9849**. This result indicates that the model can accurately distinguish between normal operations, pre-failure anomalies, and failure events with high reliability. Based on this performance, Random Forest was selected as the best initial predictive model.

Task 5: Model Optimization & Validation

Following the initial training phase, the model was further refined through advanced hyperparameter tuning and validation techniques. Cross-validation was used to improve generalization and reduce overfitting risks. Additional ensemble learning strategies were explored to combine the strengths of multiple tree-based models. The final optimized solution was an **ensemble model combining Random Forest and Extra Trees classifiers**, which achieved the best predictive performance. The optimized Random Forest achieved a **weighted F1 score of 0.9847**, while the ensemble model slightly improved performance to **0.9853**. These results demonstrate strong predictive capability and robustness in identifying early signs of asset failure in power grid systems.

Task 6: Asset Health Regime (Market Mood Index) Definition

In the final stage, the machine learning predictions were transformed into an interpretable **Asset Health Index** that provides a continuous measure of asset condition. This index ranges from **0 to 100**, where higher values represent healthier operating conditions and lower values indicate increased failure risk. The index is calculated using predicted failure probabilities combined with model confidence scores and then smoothed using exponential moving averages to reduce short-term fluctuations. Based on the index values, assets are categorized into three operational regimes: **Normal**, **Warning**, and **Critical**. For example, assets with high health scores remain in the Normal regime, while sustained declines in the index trigger Warning or Critical states. This regime-based framework enables operators to quickly identify at-risk equipment and prioritize maintenance activities.

Overall Conclusion

The predictive maintenance pipeline successfully demonstrates how machine learning and time-series analytics can transform raw power grid telemetry into actionable asset health insights. By integrating data ingestion, feature engineering, outage analysis, predictive modeling, and regime classification, the system provides a scalable framework for early detection of equipment failures. The final ensemble model achieved a **weighted F1 score of approximately 0.985**, indicating strong predictive performance. Additionally, the Asset Health Index and regime classification provide an intuitive monitoring tool that can help power utilities proactively manage infrastructure health and reduce unexpected outages.